

XPS Processing Notes — Sulfur 2p Region

Analytical chemistry group, internal reference

Smoothing and background

- At each end of the scan, extend the trace by mirroring it back across the boundary without repeating the boundary point itself: the padding value one step beyond the first recorded point equals the second recorded point, two steps beyond equals the third, and so on. The same rule applies at the other end. This gives every recorded binding energy a smoothed value.
- Take the background as the straight line through two points: the mean binding energy and mean intensity of the first five smoothed points, and the mean binding energy and mean intensity of the last five smoothed points. Subtract this line from the smoothed trace before matching references.

Reference matching

- Match each reference binding energy to the nearest local maximum of the smoothed trace (before background subtraction) within a 0.5 eV search radius.
- Where two candidate maxima are equally close to the reference, take the taller of the two.
- If no maximum falls inside the search radius, record the species as not resolved rather than forcing a match.

Censored and missing values

- Treat any concentration reported below the instrument's detection limit as half of that limit.
- Exclude a sampling day from a mean or removal calculation if it is missing either its feed or its treated-effluent value.

Reporting conventions

- Report a period's removal as calculated from that period's mean feed and mean treated-effluent concentration, not as an average of daily removals.
- Report the amplitude-versus-model gap as measured amplitude minus modelled height.
- When asked which species' modelled height and measured amplitude disagree most, compare them in absolute terms, not as a percentage.
- When comparing reduced and oxidised sulfur by modelled height, use each group's combined (summed) modelled height, not a per-component average.

Reporting precision

- Report concentrations to three significant figures and removal percentages to one decimal place.
- Report binding-energy positions and offsets to one decimal place.
- Report peak amplitudes and the amplitude-minus-modelled-height difference to one decimal place.